

Athlete Autonomy and Wearable Technology: Collegiate Athletes' Understanding and Perceptions of the Proprietary Use of Their Physical Performance Data

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Within the institutional setting of collegiate athletics in the United States, wearable technology devices are regularly issued to competing college athletes for the purpose of harvesting their personal biometric performance data. Although data harvested from college athletes by institutionally issued wearable technologies are of potential value from a performance standpoint, a robust third-party marketplace also exists for athletes' biometric performance data. At present, and in accordance with intensifying financial demands in collegiate athletics, the National Collegiate Athletic Association permits the unilateral harvest and sale of college athletes' personal biometric data. Accordingly, the present study sought to explore current college athletes' perceptions of the proprietary usage and sale of their personal biometric data. Findings illuminate both the necessity of informed governance with regard to protecting athletes' privacy amid novel technological integration and legal and financial considerations given college athletes' ability to monetize their name, image, and likeness.

In a global society increasingly reliant on data acquisition, athletics personnel (e.g., athletes, coaches, and trainers) place a premium on biometric performance data when formulating training plans, developing competitive schedules, or making programmatic decisions. Within the institutional setting of collegiate athletics in the United States, coaches and trainers have utilized wearable technologies to enhance performance and monitor training patterns of athletes (Guskey, 2022; Rudner, 2022). In fact, given athlete safety and well-being are fundamental principles in the operation of the National Collegiate Athletic Association (NCAA, n.d.), wearable technologies present an opportunity for coaches and support staff to implement more informed training regimens for the express benefit of participating athletes (e.g., load management, injury prevention; Bourdon et al., 2017; Cardinale & Varley, 2017; Luczak et al., 2020). However, the proprietary use of athletes' biometric data has faced scrutiny as the NCAA and its member institutions have sought to commodify data harvested by wearable technologies.

As wearable technologies continue to develop and integrate within collegiate athletics, biometric data gathered by such wearable technology devices have become increasingly valuable as well. For instance, apparel companies engaged in brand partnerships with NCAA member institutions have adopted contractual language that affords them near-unilateral access to biometric data gathered on institutionally issued wearable technology devices (Edelman & Holden, 2021; Kish, 2016; Tracy, 2016). In addition, the NCAA permits institutions and athletic conferences to package and sell college athletes' biometric data to sportsbooks and gambling companies (Christovich, 2022). Regulations permitting the monetization of athlete data are of specific value to sportsbooks as the gambling

industry has become increasingly reliant on technology and data to set betting lines (Edelman & Holden, 2021). While the NCAA currently has no formal policy regarding the sale of college athletes' personal biometric data, increasing corporate demands for data serve to further commodify competing college athletes.

The monetization of biometric data commodifies college athletes beyond competitive settings as training patterns, schedules, and practice performance metrics serve as potential revenue streams for NCAA member institutions. While holding direct economic value, tracking athletes' biometric data indirectly benefits NCAA athletic teams over time as well. During institutionally sponsored athletic camps or clinics, for example, it has become standard practice to utilize wearable technology devices to track prospective recruits' performance (Rudner, 2022). Given the relationship between successful athletics recruiting and athletic success (i.e., winning; Caro, 2012; Mankin et al., 2021) and the corresponding relationship between athletic success and revenue generation (Caro, 2012), the use of wearable technology within the talent evaluation process is of innate financial interest to collegiate athletics programs.

While extant research has addressed the potential legal implications of the collection of college athletes' biometric data (e.g., Arnold & Sade, 2017; Brown & Brown, 2021; Jessop & Baker, 2019; Seshadri et al., 2019; Smolenski, 2019; Studnicka, 2020), there is a sparsity of research examining college athletes' perceptions of wearable technology and the monetization of their proprietary data. Even amid criticisms related to functionality and effectiveness, NCAA member institutions continue to allocate significant resources to wearable technology, while coaches rely heavily on performance data metrics in the implementation of training regimens (Luczak et al., 2020). Given this reliance on wearable technology and the NCAA's position permitting the widespread sale of data, it stands to reason that the increasing supply of elite athletes' performance data will correspond with corporate demand (Edelman & Holden, 2021; Kish, 2016; Tracy, 2016).

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In a global economy positioning data as a form of invaluable currency (Makridis & Thayer, 2023), corporate demand for athletes' data continues to increase (Brown & Brown, 2024; Gale, 2016). Within a theoretical framework characterizing the NCAA as a hegemonic total institution, in which athletes spontaneously consent to pertinent stakeholders (e.g., coaches, administrators; Harry, 2023; Southall & Staurowsky, 2013; Southall & Weiler, 2014), this exploratory study sought to examine college athletes' perceptions of the implementation of wearable performance tracking technology and the sale of their personal biometric data.

Literature Review

Wearable Technology in Collegiate Athletics

The use of wearable technology in sport and athletics has expanded rapidly over the past decade, reshaping how athletic performance, health, and recovery are both tracked and managed. Wearable technologies (e.g., GPS trackers, heart rate monitors, accelerometers, and biometric sensors) collect real-time data on a range of physiological and biomechanical indicators (Bourdon et al., 2017; Cardinale & Varley, 2017; Luczak et al., 2020). These technologies have become particularly prominent in elite athletic settings, where athlete performance optimization and injury prevention are pivotal. Beyond performance considerations, the implementation of wearable technology introduces pressing ethical concerns regarding individual privacy, consent, and data commodification (Karkazis & Fishman, 2017).

In fact, extant research illustrates that athletes utilizing team-issued wearable technology devices often lack full knowledge of how their biometric data are collected, used, or stored, raising concerns about informed consent and agency (Corr et al., 2025; Jones et al., 2022). Such concerns can be exacerbated by the structural power dynamics that often exist in sport and athletic team settings. Accordingly, athletes may feel compelled to comply with technological monitoring due to hierarchical pressures within athletic departments, even when they have reservations about data collection practices (McMahon & Penney, 2013). Institutional partnerships with private technology firms further complicate this dynamic, as data generated by athletes can be commercialized to secure sponsorships or improve institutional/organizational branding without direct benefit to or control by athletes themselves (Brown & Brown, 2021; Jessop & Baker, 2019; Steinbach, 2022).

Such trends reflect the growing influence of data-driven governance in sport and reveal how wearable technologies serve not only as tools for health and performance monitoring but also as potential mechanisms for surveillance and control. Within collegiate athletics, biometric surveillance practices do not occur in a vacuum. Rather, they are embedded within broader institutional arrangements and ideological frameworks that shape how athlete bodies are managed, valued, and commodified (Corr et al., 2025). As such, the use of wearable technology within collegiate athletics must be understood not only through its technical functions but also in relation to the logics that dictate behavior within the institutional setting of collegiate athletics.

(Mis)Use of Wearable Technology in Collegiate Athletics

Various instances of the institutionally mandated (mis)use of wearable technology and biometric data for unsavory and abusive

purposes underscore the power dynamic college athletes compete, study, and live within. During the 2018–2019 season, for instance, women's basketball players at Texas Tech University (Texas Tech) were summarily disciplined if their heart rate fell below 90% capacity for more than 2 min during games. Mandated to wear team-issued heart rate monitors, players whose heart rate fell below this 90% threshold were subjected to intense physical conditioning and routinely received less playing time in subsequent games. A former Texas Tech player described the use of wearable technology throughout the 2018–2019 season as “a torture mechanism . . . the system wasn't supposed to be used how [coaches were] using it” (Epstein & Libit, 2020, para. 5). Other notable instances of the (mis)use of wearable technology include coaches and staff tracking college athletes' whereabouts, social behaviors, and sexual activity (Corr et al., 2025; McQuade, 2017). Such examples exemplify the stark dichotomy between institutional power and athlete autonomy and, most saliently to this research, the role and impact of wearable technology within this dynamic. As Bernstein (2020) noted, the (mis)use of wearable technology for punitive and abusive purposes will likely increase in the absence of adequate player representation and formal contractual relationships between institutions and college athletes.

Commodification of College Athletes' Biometric Data

Within the NCAA, particularly in Division I and the Football Bowl Subdivision (FBS), the dual and often conflicting institutional logics of amateurism and commercialism define the operations of athletic departments. While the NCAA promotes an amateur ideal rooted in education and personal development (Macaulay et al., 2022; Nite & Washington, 2024), a dominant commercial logic increasingly dictates policy and practice, especially at the highest competitive level of the NCAA (i.e., Power 4 conferences; Corr et al., 2022; Southall et al., 2023). Correspondingly, college athletes operate within a commercialized environment that values athletic ability and performance as a commodified package that draws significant financial interest from internal and external stakeholders (Hughes et al., 2019; Stewart, 1987). In this hyper-commercialized environment, the pursuit of revenue through corporate means often overshadows holistic developmental goals (Corr et al., 2024).

As NCAA member institutions operate within a setting that places a premium on revenue generation (Corr et al., 2022; Southall et al., 2023), the use of wearable technology to collect and sell biometric performance data from competing college athletes is not necessarily surprising. However, such emphasis on technological adaptation for revenue-generating purposes is noteworthy given the ethical, legal, and privacy concerns surrounding wearable technology in collegiate athletics (e.g., Arnold & Sade, 2017; Brown & Brown, 2021; Jessop & Baker, 2019; Seshadri et al., 2019; Smolenski, 2019; Studnicka, 2020). In addition, in the name, image, and likeness (NIL) era, athletic departments' proprietary use of performance data may disproportionately serve to the financial detriment of college athletes. As user data are a premium commodity in the modern global economy (Makridis & Thayer, 2023), the collection and sale of athletes' biometric data without informed consent faces legal scrutiny. In fact, Jessop (2022) contended that the mandatory usage of wearable technology in collegiate athletics violates college athletes' right to privacy, as granted by the Fourth Amendment of the United States Constitution, and personal property law.

Theoretical Framework

Collegiate Athletics as a Total Institution

Drawing on Goffman's (1961) concept of the total institution, collegiate athletics in the United States can be conceptualized as a system in which the boundaries between college athletes' educational, social, and athletic pursuits are circumscribed by institutional imperatives. Within this setting, competing college athletes experience highly structured daily routines, limited spatial mobility, and hierarchical power relationships that shape their identities, behaviors, and perceptions of agency (Hatteberg, 2018). The physical, temporal, and emotional demands placed on college athletes mirror those of total institutions, where individuals are removed from the broader social world and subjected to all-encompassing control mechanisms. Notably, however, the authors do wish to delineate between the application of the total institution framework in traditional settings (e.g., prison and military) and within the context of collegiate athletics. While the application of the total institution setting is apt within collegiate athletics (Corr & Hoffman, 2025; Stokowski et al., 2023), the setting is nonetheless *extremely* privileged in comparison to that of settings where confinement and/or autonomous decision making are oppressed to a significantly greater extent.

Institutional Hegemony and Power Relations

Scholars have argued that collegiate athletics constitutes a hegemonic institution characterized by entrenched power asymmetries and patriarchal social arrangements (Corr & Hoffman, 2025; Harry, 2023; Southall & Weiler, 2014). Within this structure, college athletes are positioned as subordinate actors whose time, behaviors, and labor are managed by institutional agents (e.g., coaches, administrators, boosters, and media stakeholders). These dynamics are codified through both formal policies (e.g., team handbooks and department codes of conduct) and informal cultural norms that normalize the suppression of college athletes' agency (Springer et al., 2025).

Within this setting, coaches often embody surrogate parental roles, exerting control over athletes' routines and social behaviors (Corr et al., 2023; Stokowski et al., 2023). Such paternalistic authority not only reinforces compliance but also aligns with broader institutional goals, fostering a culture in which dissent is minimized and institutional priorities go unchallenged (Kluch, 2023). The result is an environment that discourages opposition and positions athlete subordination as necessary for personal and team success. Despite operating under the logic of amateurism, many athletes have internalized what Macaulay et al. (2022) described as a professional-amateur logic (i.e., an ideological stance that simultaneously accepts amateur status while rationalizing the need for institutional revenue generation). This form of spontaneous consent reveals the powerful cultural work done by hegemonic institutions to secure compliance through normalization rather than overt coercion.

Athlete Oversight and Surveillance

The integration of wearable technology into collegiate athletics further exemplifies the controlling and commodifying tendencies of the institution itself. While such technologies are publicly promoted as tools for performance enhancement and injury prevention (Bourdon et al., 2017; Cardinale & Varley, 2017; Luczak

et al., 2020), they also serve commercial and surveillance functions. Partnerships with technology firms (e.g., Penn State's collaboration with WHOOP) enable institutions to collect college athletes' biometric data (e.g., heart rate, sleep patterns, and recovery metrics) under the guise of holistic wellness. However, many athletic departments also utilize such personal biometric data for competitive and financial gain (Corr et al., 2025; Penn State Athletics, 2023; Steinbach, 2022). These practices raise critical ethical concerns about privacy, data ownership, and informed consent (Corr et al., 2025; Jones et al., 2022).

Given the existing power dynamics in athlete-coach and athlete-institution relationships, college athletes often lack both the knowledge of and control over how their data are collected, used, or sold. Although federal and state laws (e.g., Family Educational Rights and Privacy Act) provide partial protections, the lack of NCAA policy permits the monetization of college athletes' personal biometric data without athlete compensation or clear opt-out pathways (Brown & Brown, 2021; Jessop & Baker, 2019). The institutional enforcement of biometric surveillance is further facilitated by the paternalistic underpinnings of the athlete-coach dynamic, normalizing mandated wearable technology adoption as a condition of participation. In this way, wearable technologies become disciplinary tools within the total institution itself, a tool that simultaneously serves institutional goals and strips athletes of autonomy over their own bodies.

Embedded in the contradictions between amateurism and commercialism (Macaulay et al., 2022), athletes navigate a socio-cultural and organizational landscape defined by control, commodification, and surveillance. While positioned as students, they are increasingly treated as laborers whose bodies, time, and data are managed in service of institutional and corporate interests (Southall et al., 2023). Accordingly, the combination of total institutional setting in which college athletes exist and extant research indicating college athletes may be uneducated or unaware of how personal biometric data are collected, aggregated, and/or sold (Corr et al., 2025; Jones et al., 2022), the present study sought to explore current college athletes' perceptions of the usage and sale of their personal biometric data.

Methodology

Disruptive Research Approach

Within critical sociology, scholars are not passive observers but active participants in challenging the structural inequities embedded in systems, such as the NCAA. Rooted in a critical paradigm, this methodological approach views research as a tool for praxis, not just to understand the world but to change it (Kincheloe & McLaren, 2011). Burawoy's (1982) notion of public sociology calls on researchers to elevate marginalized voices, disrupt dominant ideologies, and engage society in democratic debate. In this view, knowledge production is inherently political and should support collective consciousness-raising. This paradigm prioritizes reflexive, participatory methodologies that center participants' agency while interrogating institutional power (Sparkes, 2002). Scholars challenge myths like amateurism and meritocracy by naming the ideological foundations that normalize exploitation. Accordingly, research becomes a strategic intervention and a means of surfacing hidden structures, disrupting cognitive frames, and contributing to emancipatory social change. In doing so, scholars illuminate what institutions obscure, supporting transformation from within and beyond the academy.

In the context of this work, the researchers occupied the role of *disruptors* among participants currently competing in collegiate athletics. As the findings below illustrate, many participants were unaware that the NCAA permitted the package and sale of college athletes' personal biometric data gathered from institutionally issued wearable technology devices. In this sense, the researchers actively informed participants of practices permitted by the NCAA and then gathered their immediate reactions to the presentation of this information. As nearly all participants indicated that college athletes should retain the rights to their personal biometric data—regardless of source of collection (i.e., institutionally or personally)—the researchers' role in communicating these data to participants was not exclusively proprietary, representing a disruptive factor within this space. In addition, in conveying the potential value of college athletes' biometric data to corporate entities (e.g., apparel companies and sportsbooks), the researchers actively highlighted a potential revenue source unknown to participants prior to partaking in this study. While the viability of monetizing one's individual biometric data is undoubtedly marginal, the role the researchers occupied within this study exemplifies the opportunity for scholars to disseminate information and, potentially, disrupt distinct institutional settings.

Research Design

Given that this research was exploratory in nature, the researchers employed semistructured interviews with current NCAA athletes to investigate their perceptions of wearable technology in collegiate athletics. The primary aim of this study was to explore competing college athletes' perceptions of the proprietary usage and sale of college athletes' personal biometric data. The researchers conducted this study as a follow-up to previous research exploring the function of wearable technology in collegiate athletics (Corr et al., 2025). Whereas this previous research sought to explore the function of wearable technology in collegiate athletics, this research explicitly concerned college athletes' perceptions of the proprietary usage and sale of their biometric data. Researchers also asked probing questions to gather responses pertaining to participants' sentiments regarding biometric data rights and the monetization of biometric data. The complete interview guide used in this study can be referenced in the [Appendix](#).

Interviews lasted between 20 and 33 min ($M = 26$ min) and were recorded and stored on a secure audio device only accessible to the researchers. Two researchers conducted all interviews in accordance with a consistent semistructured interview protocol. Interviews with participants were conducted in the spring of 2024 and conducted remotely via the virtual meeting application, *Zoom*. As this study was conducted as a follow-up study and specifically aimed to explore athletes' perceptions of the proprietary usage and sale of athletes' biometric data, the researchers focused exclusively on questions about biometric data usage and sale.

Participants

The study involved 54 NCAA athletes ($n = 54$) actively competing in collegiate athletics during the 2023–2024 academic year. At the time interviews were conducted, all of the participants competed at NCAA Division I institutions. Participants were recruited according to their participation in a preliminary study conducted by the researchers in the fall of 2023 (Corr et al., 2025). While this initial study included 59 participants, the researchers limited it to athletes competing at the Division I level. Such a decision was made due to

the fact that the 2022 NCAA decision to ease restrictions on data sharing with third parties was made by the Division I Interpretations Committee and applied to Division I member institutions (Christovich, 2022; Novy-Williams, 2022).

Having participated in a previous study in the fall of 2023, participants were contacted via email or text message in February 2024 to gauge their willingness to participate in an additional interview. The two researchers who facilitated each of the interviews contacted each of the participants directly. All 54 of the contacted participants agreed to take part in a follow-up interview. Interviews were scheduled around participants' competition, training, and class schedules in February and March 2024, typically taking place in the evening. The 54 participants represented nine distinct NCAA sports. Of these, 36 were women, and 18 were men. Table 1 presents anonymized demographic information for each participant.

Data Analysis

Once interviews were completed, the researchers transcribed the audio recordings and sent the transcripts to participants for review, allowing them to confirm or clarify their responses (i.e., member checking; Harvey, 2015). After gathering verified transcripts, researchers began an inductive coding process. Using participants' exact wording, an *in vivo* coding method was employed to extract key themes from the interviews. Given the exploratory nature of this study, an inductive coding process that derived direct meaning from participants' comments was utilized to create corresponding themes. Axial coding was undertaken during the second round of coding to group related *in vivo* themes (Creswell et al., 2007). Throughout the process, the researchers cross-checked their notes to ensure consistency in theme identification and establish a measure of validity (Thompson et al., 1989). Pseudonyms were assigned to participants, and all identifying details were removed to maintain confidentiality.

During these rounds of coding, the research team collaborated to identify key thematic areas that emerged from the data. Given the researchers' implementation of an inductive coding schema, the thematic areas were based specifically on group *in vivo* codes. The emergent thematic categories identified by the researchers concerned: (a) consent to biometric tracking, (b) perceptions of the sale of biometric data, and (c) athletes' right to personal biometric data. These themes were derived from the responses and, subsequently, a round of coding. Accordingly, the findings section is structured around these themes.

Trustworthiness and Reflexivity

It is crucial for researchers analyzing interview data to practice reflexivity during analysis (Mauthner & Doucet, 2003). To maintain transparency, the researchers openly discussed their own positionality and personal experiences as part of this reflective practice. Each researcher holds a faculty position in a sports-related academic field and has substantial professional experience in collegiate athletics. Two of the researchers previously competed in varsity sports, while two others are currently involved in athlete development at their respective universities. The researchers' backgrounds in athletics and their firsthand experiences interacting with college athletes led them to develop the focus of this exploratory study. The researchers collaborated closely during the coding process, regularly comparing notes to achieve consensus on emergent themes.

Table 1 Characteristics of Participants

Pseudonym	Division I classification	Gender	Sport	Signed informed consent?
Abigail	No football	Women's	Tennis	No
Amelia	No football	Women's	Soccer	No
Aria	No football	Women's	Tennis	No
Aurora	FBS	Women's	Soccer	No
Ava	No football	Women's	Tennis	No
Avery	FBS	Women's	Volleyball	No
Benjamin	FBS	Men's	Football	No
Camila	FBS	Women's	Soccer	No
Charlotte	FBS	Women's	Track and field	No
Chloe	FCS	Women's	XC, track and field	No
Daniel	FBS	Men's	Track and field	Yes
Eleanor	FBS	Women's	Track and field	No
Eliana	No football	Women's	Tennis	No
Elijah	FCS	Men's	Track and field	No
Elizabeth	No football	Women's	Soccer	No
Ella	FCS	Women's	Tennis	No
Ellie	FBS	Women's	Tennis	No
Emily	No football	Women's	Soccer	No
Emma	No football	Women's	Soccer	No
Evelyn	No football	Women's	Soccer	No
Ezra	FBS	Men's	Football	No
Gianna	No football	Women's	Soccer	No
Harper	No football	Women's	Soccer	No
Hazel	FBS	Women's	Tennis	No
Henry	No football	Men's	Tennis	No
Isabella	No football	Women's	Soccer	No
Isla	No football	Women's	Tennis	No
Ivy	No football	Women's	Tennis	Yes
Jack	FBS	Men's	Football	No
James	No football	Men's	Track and field	No
Layla	No football	Women's	Soccer	No
Levi	No football	Men's	Track and field	No
Liam	No football	Men's	Track and field	No
Lily	No football	Women's	Tennis	No
Lucas	FBS	Men's	Football	No
Lucy	FBS	Women's	Soccer	No
Luna	FBS	Women's	Tennis	No
Mateo	FBS	Men's	Football	No
Mia	FBS	Women's	XC, track and field	No
Michael	FBS	Men's	Football	Yes
Mila	No football	Women's	Track and field	No
Noah	FBS	Men's	Football	No
Nora	No football	Women's	Soccer	No
Oliver	FBS	Men's	Football	No
Olivia	FBS	Women's	Rowing	No
Penelope	No football	Women's	Soccer	No
Samuel	FBS	Men's	Track and field	No
Scarlett	No football	Women's	Tennis	No
Sebastian	FCS	Men's	Football	No
Sophia	FBS	Women's	Track and field	No

*(continued)***(Ahead of Print)**

Table 1 (continued)

Pseudonym	Division I classification	Gender	Sport	Signed informed consent?
Theodore	FBS	Men's	Track and field	No
Violet	No football	Women's	Soccer	Yes
William	No football	Men's	Track and field	No
Zoe	No football	Women's	Track and field	No

Note. FBS = Football Bowl Subdivision; FCS = Football Championship Subdivision; XC = cross country.

Findings

Consent to Biometric Tracking

While many athletes indicated their spontaneous consent to biometric tracking in and out of athletic performance settings, only four participants (7%) affirmed they signed a tangible consent form prior to being tracked by athletic department institutional members. Ivy, a women's tennis player, indicated that athletics compliance personnel had issued consent forms prior to her and her teammates being issued wearable technology. She stated, "I understand the usage and have been informed through compliance. I permit the sharing of my information." Michael, an FBS football player, had also signed an informed consent document permitting his collected biometric data to be stored and sold by the athletic department. He stated he would be uncomfortable with such practice "if no informed consent is given by the athlete." Notably, each of the four participants who had signed an informed consent document stated they believed college athletes should retain the rights to their biometric data.

Although these four participants had knowingly signed a document consenting to the collection, storage, and sale of their biometric data, the vast majority of participants ($n = 50$) had not signed informed consent documents or been educated on the usage and potential sale of their personal biometric data by their respective institutions. Ellie, a women's tennis player, summed up the feeling of most participants regarding the lack of consent prior to being issued university-owned wearable technology, stating, "I didn't provide informed consent to the sale of my data, unless it was in very fine print that I didn't read on back-to-school forms. So that is upsetting." In addition, 35 participating athletes (65%) were unaware that the NCAA permits the collection and sale of athletes' biometric data to third-party companies such as sports apparel companies or sportsbooks. In some cases, participants indicated they did not know if they consented to the collection and sale of biometric data. For instance, Chloe, a women's cross-country and track-and-field runner, stated as much, saying, "I don't have time to read all the forms they make us sign so I wouldn't be surprised if I had agreed to let them sell the data at some point."

Regardless of previous awareness of institutional permissions to monetize athlete data, the majority of participating athletes expressed discomfort over the lack of autonomy they possessed over their personal biometric data. Layla, a women's soccer player, stated, "It's hard because using the wearable technology is like a non-negotiable. I'd like my health data to stay between the staff and my AT [athletic trainer]. This isn't communicated that this data is sold." Benjamin, an FBS football player, emphasized the importance of awareness concerning the sale of athletes' biometric data: "I personally think athletes should be asked or warned before sending their information off." Olivia, a women's rower, noted the importance of consent, stating, "I think informed consent is important here. Having athletes sign contracts without taking the

time to educate them on what they're signing. [Education] should be the priority." Aurora, a women's soccer player, stated similarly, "I think we should be aware of it and sign a release to say that we agree that our data can be sold." Although he was uncomfortable with the sale of college athletes' biometric data, Daniel, a men's track-and-field athlete, was not surprised to learn the NCAA permitted this type of transaction, noting, "it's a pure business." Ellie stated more succinctly, "We are their product, so they own us. Isn't that what an NLI [National Letter of Intent] signature is?"

Perceptions of the Sale of Biometric Data

Consistent with a theoretical framework characterizing the NCAA as a hyper-commercialized institutional setting in which college athletes are hyper-commodified components, multiple responses regarding the sale of athletes' biometric data alluded to the lucrative marketplace college athletes compete in. William, a men's track-and-field athlete, stated, "It has always felt like a dog show to be honest. And to learn they exploit us even more than they let on—they let us know a lot too—is not the best feeling in the world." Oliver, an FBS football player, codified the sale of athletes' biometric data within the lucrative NCAA marketplace:

I guess it's the nature of the game. Even college athletics, an amateur sport, has been hyper-commercialized. But now with NIL [name, image, and likeness] being a thing, any athlete that has their data sold should also see a percentage of the payout. And I know companies in any field pay top dollar for data. I think it's fair compensation for the physical demands that come with the micromanaging of an athlete's body.

Interestingly, athletes' perceptions and understanding of the NIL marketplace appeared to be a focal point of many participant responses. When prompted to describe her feelings related to the sale of athletes' biometric data, Olivia stated, "I think it's unfair that [athletic programs] can profit off of athletes' information, even if the data is anonymous." Elijah, a men's track-and-field athlete, stated similarly, "I would not permit [the sale of biometric data] unless I got the benefits, such as money, from these transactions." Samuel, a men's track-and-field athlete, also said athletes should retain the use of their biometric data: "That is information from your body. You have the right to not have the university use pictures of yourself so why not have the right to not have your biometrics sold?" Seemingly acutely aware of the potential market value of college athletes' biometric data, many participants were uncomfortable with the NCAA's expressed permission for athletic departments and/or conferences to package biometric data for sale to third-party companies without compensating athletes directly.

While many athletes were uncomfortable with the sale of their biometric data, a handful expressed little to no concern over the monetization of college athletes' data. In fact, a number of athletes focused on the benefits of readily sharing biometric performance data with third parties. Charlotte, a women's track-and-field athlete,

detailed the benefit apparel companies could see from possessing college athletes' biometric data, stating, "I think it can help these companies improve their products and therefore improve the experience of student-athletes in the future." Evelyn, a women's soccer player, believed apparel companies holding college athletes' biometric data gave them "the info they need to target the needs of college athletes." Levi, a men's track-and-field athlete, stated that the sale of biometric data to apparel companies "could help improve the clothes we wear, to shoes we run in, or the workouts we go through every day." Liam, a men's track-and-field athlete, stated that sharing biometric data with apparel companies was "Great. A good way to show the products are working with the highest athletes." Theodore, also a men's track-and-field athlete, indicated that the purpose of shared and sold biometric data was necessary to consider. He stated, "I feel it is ok to use or sell data in order to provide specifics or information for scientific studies. But I do not approve selling [biometric data] to apparel companies just to generate a greater profit."

Such sentiments regarding the dissemination or sale of college athletes' biometric data to apparel companies or research studies were common among many participants. However, opinions differed drastically when discussing the sale of college athletes' biometric data to gambling entities such as sportsbooks. Sebastian, a Football Championship Subdivision football player, discussed the dichotomy between transaction types:

Apparel companies I understand. These companies could use that data to enhance certain fabrics and the technology inside their gear. Permitting the sale of data to sportsbooks is wild. My question would be why do they need that information? I believe that it would give an advantage to bettors in an already corrupt gambling system.

Lucas, an FBS football player, also questioned the specific sale of college athletes' biometric data to sportsbooks, stating "I think that is an inherent conflict of interest and I'm sure most athletes don't realize they are consenting to that."

Right to Personal Biometric Data

Regarding the retention of autonomous control over one's biometric data, participants nearly unanimously agreed that college athletes should retain the rights to their biometric data. Ava, a women's tennis player, focused on the characterization of biometric data itself, stating, "They need to tell athletes about this before they tell them to wear these things. They need to have the choice to opt out of sharing their data. I don't understand the need [to sell data]." Chloe added, "I don't have a choice on wearing my watch to practice so I should at least have a choice on what my coach [or] school does with the data." Scarlett, a women's tennis player, stated more bluntly, "I have no option." Ellie pushed back on the commodification of college athletes, stating "Bodily autonomy is regularly forgotten in college athletics. Your data is your data. You are a person, not a product." However, it is worth noting that three participants indicated they were willing to cede control of their biometric data. James, a men's track-and-field athlete, justified his willingness to relinquish control of his personal biometric data:

College athletics is a business, and their source of income is us as athletes. They are paying for our school so as long as it doesn't negatively impact my privacy, I think they can do what they want with our data.

While James and two other participants (Amelia and Nora) indicated that they were willing to relinquish control of their biometric data, the overwhelming majority of athletes believed that personal biometric data are the property of the user themselves and no one else. Ava summed up succinctly:

It's my body and my data. I should be able to use that how I want to personally use it. If coaches, nutritionists, trainers, et cetera believe it's something worth sharing or talking about, then they need to ask me first and pitch their reasoning. There could be many different factors to why I am the way I am, and I'm more than a heart rate or body composition.

Discussion and Implications

NCAA member institutions operate within a hyper-commercialized setting that positions athletics-based revenue generation as of utmost importance (Corr et al., 2022; Southall et al., 2023). Accordingly, the monetization of athlete biometric data collected from wearable technologies is consistent with commercial efforts in NCAA athletics. However, most athletes in this study (65%) indicated they were unaware that their unique biometric data were able to be packaged and sold. While such data deals have not been made explicitly public, the NCAA (2023) clarification to rules and regulations pertaining to the sale of athlete biometric data is indicative of the perceived need to provide clarity on the matter of proprietary usage of biometric performance data.

The findings of this study also reflect at least partial awareness among participants of the commodification of college athletes. While the commercial interests of stakeholder groups within the collegiate athletics enterprise stimulate such commodification, the sale of college athletes' personal biometric data exemplifies the hyper-commercialized logic pervasive in collegiate athletics (Corr et al., 2022; Southall et al., 2023). Given that wearable technology devices capturing athletes' biometric data are issued by members of an athletic department, such a process is institutionalized. As the power dynamics in collegiate athletics reflect a hegemonic orientation—in which spontaneous consent is given and questioning the decisions of superiors is discouraged—college athletes face little in the way of recourse to retain the rights to their personal biometric data. Although spontaneous consent is a central tenet of hegemonic theory (Gramsci, 1971), participants' responses in this study indicate college athletes may indeed question the varying standardized and institutionalized practices they are subjected to. While college athletes may tacitly acquiesce to institutional policies and expectations, they do not necessarily do so enthusiastically or unconsciously (Stokowski et al., 2025). Scarlett's response of "I have no option," and Ellie's comment of "it's just what it is when we get [to college]" with regard to the collection, usage, and sale of her biometric data are indicative of such acquiescence. However, the vast majority of participants in this study (94%, $n = 51$) stated that college athletes should be the sole proprietors and financial beneficiaries of their personal biometric data.

Coinciding with the findings of this study, college athletes are seemingly uninformed about the proprietary usage and sale of their biometric data and the nature in which wearable technologies are becoming ever pervasive within their day-to-day lives as competing college athletes. Under the guise of protecting and developing athletes, coaches and athletic department staff routinely monitor athletes' everyday activities via wearable technology devices (Corr et al., 2025). Phrased as benevolent, the power dynamics of collegiate athletics—in which competing athletes have marginal

representation or form of recourse—manifest in a system that limits athletes' autonomous decision making as deferential to that of their respective coach. In accordance with the findings of this study, regardless of athletes' perceptions of biometric data usage and sale, current college athletes hold little ability to functionally alter the implementation of wearable technology.

This coercive setting in which college athletes exist lends itself to privacy concerns regarding the ownership of athlete biometric data. Given the annual fragility of athletic grant-in-aid agreements (i.e., athletic scholarships) and the absence of formal protection (e.g., unionized body and players' association), many college athletes are fearful of retribution on behalf of their coaching staff or administration (Dzieciaszek & Mugford, 2018). The power structure of NCAA athletics centers head coaches and coaching staffs as holding singular authoritative control in which athletes are expected to spontaneously consent (i.e., hegemonic control; Harry, 2023; Hatteberg, 2018). Accordingly, many college athletes—as well as prospective recruits—experience unique pressures to remain complicit within the system they exist for fear of reprisal (e.g., reduced playing time, punitive conditioning, and loss of grant-in-aid; Alexander et al., 2024; Stokowski et al., 2023). In the context of wearable technologies, while college athletes may feel uncomfortable with the collection, dissemination, and sale of their data, they are unlikely to resist the implementation of biometric monitoring regimens.

The findings of this study also coincide with a total institution metaphor, in which college athletes' lives are overtly structured by and around their respective athletic departments (Hatteberg, 2018; Southall & Weiler, 2014). Consistent with this metaphor, coaches monitor and script players' lives in accordance with the strategic objectives of the team and institution itself (Keefer, 2023). Given the mandated use of wearable technology during performance and personal settings, coaches have the ability to routinely monitor athletes through their daily activities, creating a disconnect between team activities and personal time (Corr et al., 2025). Within the context of the commodification of college athletes and commercialization in collegiate athletics, regimented biometric and location tracking occurs in accordance with the monetary value and investment athletic departments make in scholarship athletes. In order to reap the maximum return on this investment, athletic departments are emboldened to monitor, surveil, and invade athletes' privacy. Such actions correspond to a total institutional metaphor (Hatteberg, 2018; Southall & Weiler, 2014) and are common occurrences in highly restrictive environments, such as the military or penal systems, in which overt opposition is unconscionable. College athletes acquiesce to these practices because they expect them to occur. In the context of a paternalistic institution setting, college athletes are likely to trust that their coaches and institutions have their best interests in mind (Corr et al., 2023). Such trust manifests in college athletes being vulnerable and susceptible to manipulation.

Considering the NCAA's citation of legal precedent permitting the involuntary servitude of prisoners as grounds to restrict college athlete compensation (Livers v. NCAA, 2018), the application of a total institution metaphor is justifiable in the context of collegiate athletics. However, while the total institution metaphor is applicable, it is important to delineate the total institutional setting of collegiate athletics as one with autonomous limitations rather than direct autonomous oppression. Although a confluence of personal, sociocultural, and financial factors undoubtedly influences a competing athlete's decision to persist within this total institutional setting, the agency to discontinue participation in

collegiate athletics is strikingly different from that of other traditional total institutional settings.

Practical Recommendations

While the NCAA continues to lobby the legislative and executive branches of the federal government, the restoration of athletes' rights to control and monetize their personal biometric data should be at the fore. In the National Football League (NFL), for instance, the NFL Players' Association has partnered with wearable technology companies (e.g., WHOOP, 2020) on licenses that supply every active NFL player with wearable technology device(s). While the partnership financially benefits the NFL Players' Association itself, it is explicitly stated that players will retain ownership and sole proprietary rights to biometric data collected on wearable technology devices. In collegiate athletics, however, the NCAA's unwillingness to declare competing athletes as employees of their respective athletic departments precludes the negotiation of undeniably lucrative licensing agreements on behalf of athletes. Rather, third-party firms have collectively facilitated agreements between NCAA member institutions, college athletes, and corporate parties. When Electronic Arts (EA) reprised the EA Sports College Football video game franchise in 2024, for instance, EA signed a licensing agreement with Collegiate Licensing Company that led to compensation for included FBS football players and teams featured in the game (Nakos, 2024). Accordingly, EA Sports College Football did not feature any NCAA logos or language pertaining to the NCAA itself.

Given the 2024 reprisal of EA Sports College Football became the best-selling sports video game of all time (Vannini, 2025), the NCAA's reluctance to embrace group licensing agreements is a significant unrealized potential revenue stream. In the context of wearable technology and college athletes' biometric data, transparency and an organizational willingness to adapt could perhaps result in significant financial gains for the NCAA and its member institutions. While this shifting position would align with the commercial logic(s) undergirding the NCAA, the present and traditional orientation of the NCAA makes such an outcome unlikely.

Limitations

Given the relative lack of awareness of participating college athletes regarding the proprietary usage of their personal biometric data, interviews were informative of current NCAA policy and practice for participants. In this sense, the researchers served as hegemonic disruptors for participants unaware of the proprietary usage of their personal biometric data. The researchers began each interview by asking participants if they were aware of the practice of packaging and selling college athletes' biometric data. Participants were then asked about their perception of such permissible practice and additional questions regarding the utilization of wearable technology devices. The impact of orienting the interview protocol in this manner is a necessary limitation to explicitly identify. While the purpose of this paper was to examine college athletes' perceptions of the proprietary usage and sale of their personal biometric data, participants' responses regarding the proprietary usage and control of their personal biometric data were undoubtedly influenced by the revelation that college athletes' data were permitted to be packaged and sold without their consent. As observable in the findings section of this manuscript, participants' responses routinely centered on the sale of their data, rather than

larger conversations surrounding biometric data ownership. Combined with the limited duration of interviews ($M = 26$ min), preventing further in-depth conversations concerning wearable technology usage and implementation, the sequencing of interview questions served as a notable potential limitation of this study.

Conclusion and Future Research

Within an information society comfortable with collecting information en masse from competitive athletes, the manner in which data is gathered by colleges and universities on their respective athletes is worthy of future examination. While college athletes undoubtedly perceive wearable technology to have some performance value, findings indicating the lack of education and/or consent regarding biometric data collection and sale are striking. As ancillary revenue streams continue to emerge and be prioritized within the collegiate athletics marketplace, the proprietary usage and sale of athletes' biometric data face legal scrutiny. Such scrutiny is compounded in the NIL era as college athletes are permitted to monetize their NIL.

Given corporate interest in elite athletes' biometric data (Edelman & Holden, 2021; Kish, 2016; Tracy, 2016), future research examining the utilization of college athletes' biometric data would be valuable to conceptualizing the tangible, monetized value of athlete performance data. While extant scholarship has examined the legality of tracking and storing college athletes' biometric data with or without their informed consent (Brown & Brown, 2021; Jessop & Baker, 2019), the integration of gambling and sports wagering in collegiate athletics provides an additional area for legal inquiry. As the NCAA and member conferences and athletic departments continue to explore financial partnerships with sportsbooks and gambling platforms (Christovich, 2023), the role of college athletes' biometric data within such transactions raises questions regarding gambling and sports wagering policies at the NCAA and departmental level. As sportsbooks increasingly rely on athlete performance data to set more informed betting lines (Edelman & Holden, 2021), partnerships with the NCAA and its member conferences and athletic departments may also reflect an increase in the valuation of college athletes' biometric data. Future studies examining the intersection between biometric data and information technology, data security, and risk management would be valuable from an organizational studies standpoint.

References

- Alexander, K.N., Adams, K.V., & Dorsch, T.E. (2024). "We just couldn't talk about that": Unmasking male student-athletes' experiences of emotional abuse. *The Journal of Men's Studies*, 32(3), 552–576. <https://doi.org/10.1177/10608265241245912>
- Arnold, J.F., & Sade, R.M. (2017). Wearable technologies in collegiate sports: The ethics of collecting biometric data from student-athletes. *The American Journal of Bioethics*, 17(1), 67–70. <https://doi.org/10.1080/15265161.2016.1251648>
- Bernstein, D. (2020). Texas Tech basketball abuse allegations show risk of wearable tech in sports. *The Sporting News*. <https://www.sportingnews.com/us/ncaa-women/news/texas-tech-abuse-allegations-wearable-tech-in-sports/1lan1crzql81wxloz0ibzqjr>
- Bourdon, P.C., Cardinale, M., Murray, A., Gastin, P., Kellmann, M., Varley, M.C., Gabbett, T.J., Coutts, A.J., Burgess, D.J., Gregson, W., & Cable, N.T. (2017). Monitoring athlete training loads: Consensus statement. *International Journal of Sports Physiology and Performance*, 12(2), 161–170. <https://doi.org/10.1123/IJSP.2017-0208>
- Brown, S.M., & Brown, K.M. (2021). Should your wearables be shareable? The ethics of wearable technology in collegiate athletics. *Marquette Sports Law Review*, 32(1), 97–116.
- Brown, S.M., & Brown, K.M. (2024). Betting on athlete data: The legal landscape of professional sports, athletes' rights and gaming companies. *Journal of Global Sport Management*, 9(2), 325–345. <https://doi.org/10.1080/24704067.2022.2098158>
- Burawoy, M. (1982). *Manufacturing consent: Changes in the labor process under monopoly capitalism*. University of Chicago Press.
- Cardinale, M., & Varley, M.C. (2017). Wearable training-monitoring technology: Applications, challenges, and opportunities. *International Journal of Sports Physiology and Performance*, 12(2), 55–62. <https://doi.org/10.1123/ijsp.2016-0423>
- Caro, C.A. (2012). College football success: The relationship between recruiting and winning. *International Journal of Sports Science & Coaching*, 7(1), 139–152. <https://doi.org/10.1260/1747-9541.7.1.139>
- Christovich, A. (2022). NCAA reportedly allowing sale of player data to sportsbooks. *Front Office Sports*. <https://frontofficesports.com/ncaa-reportedly-allowing-sale-of-player-data-to-sportsbooks%EFBF%BC/>
- Christovich, A. (2023). The increasingly uneasy relationship between college sports and gambling. *Front Office Sports*. <https://frontoffice.sports.com/the-increasingly-uneasy-relationship-between-college-sports-and-gambling/>
- Corr, C., & Hoffman, R. (2025). Big games, little voices: Patriarchy and power at SEC football media days. *Communication & Sport*, 10, Article 1153. <https://doi.org/10.1177/21674795251391153>
- Corr, C., Southall, C., Hawkins, B., & Southall, R.M. (2023). College football "kids": Infantilizing language in Football Bowl Subdivision bowl game broadcasts. *Sociology of Sport Journal*, 41(2), 172–179. <https://doi.org/10.1123/ssj.2022-0185>
- Corr, C., Southall, C., & Southall, R.M. (2022). The ESPNification of FBS college football: The adoption of an integrated marketing communication televisuality in FBS bowl game broadcasts. *International Journal of Sport Communication*, 15(2), 139–147. <https://doi.org/10.1123/ijsc.2021-0124>
- Corr, C., Stokowski, S., Berg, B.K., Gearity, B., & Sabin, E. (2024). "What is the ROI on that?": Examining the lack of transition support for collegiate athletes. *Journal of Organizational Psychology*, 24(3), 35–50. <https://doi.org/10.33423/jop.v24i3.7245>
- Corr, C., Stokowski, S., Staurowsky, E.J., Paule-Kobe, A.L., & Southall, R.M. (2025). The modernization of monitorization: Technological adaption and innovation as mechanisms for college athlete surveillance. *Journal of Sport & Social Issues*, 49(2–3), 175–185. <https://doi.org/10.1177/01937235251330206>
- Creswell, J.W., Hanson, W.E., Clark Plano, V.L., & Morales, A. (2007). Qualitative research designs: Selection and implementation. *The Counseling Psychologist*, 35(2), 236–264. <https://doi.org/10.1177/0011000006287390>
- Dzieciaszek, G.R., & Mugford, A. (2018). Working with organizations in sport, exercise and performance psychology. In A. Mugford & J.G. Cremades (Eds.), *Sport, exercise and performance psychology* (pp. 75–85). Routledge.
- Edelman, M., & Holden, J.T. (2021). Monopolizing sports data. *William & Mary Law Review*, 63(1), 69–135.
- Epstein, J., & Libit, D. (2020). Texas Tech women's basketball players describe toxic culture: 'Fear, anxiety and depression.' *USA Today*. <https://www.usatoday.com/in-depth/sports/ncaaw/big12/2020/08/05/marlene-stollings-texas-tech-program-culture-abuse-players-say/5553370002/>

- Gale, K. (2016). The sports industry's new power play: Athlete biometric data domination. Who owns it and what may be done with it. *Arizona State University Sports & Entertainment Law Journal*, 6(1), 7–84.
- Goffman, E. (1961). *Asylums: Essays on the social situation of mental patients and other inmates*. Anchor Books/Doubleday.
- Gramsci, A. (1971). *Selections from the prison notebooks of Antonio Gramsci*. International Publishers.
- Guskey, J. (2022). Inside Kansas football's use of the data it collects monitoring player activity each week. *The Topeka Capital-Journal*. <https://www.cjonline.com/story/sports/college/hawk-zone/2022/09/23/kansas-football-catapult-nordbord-lance-leipold-big-12-ncaa-football-sports-science/10434444002/>
- Harry, M. (2023). Hegemony and the National Collegiate Athletic Association: A critical discourse analysis of National Collegiate Athletic Association resources concerning name, image, and likeness. *Sociology of Sport Journal*, 41(2), 152–162. <https://doi.org/10.1123/ssj.2023-0017>
- Harvey, L. (2015). Beyond member-checking: A dialogic approach to the research interview. *International Journal of Research & Method in Education*, 38(1), 23–38. <https://doi.org/10.1080/1743727X.2014.914487>
- Hatberg, S.J. (2018). Under surveillance: Collegiate athletics as a total institution. *Sociology of Sport Journal*, 35(2), 149–158. <https://doi.org/10.1123/ssj.2017-0096>
- Hughes, R.P., Kimball, E.W., & Koricich, A. (2019). The dual commodification of college going: Individual and institutional influences on access and choice. In L.W. Perna (Ed.), *Higher education: Handbook of theory and research* (pp. 415–477). Springer.
- Jessop, A. (2022). Examining the Fourth Amendment and biometric data privacy law implications of NCAA athletes' mandated use of athlete biometric data tracking devices. *Cardozo Arts & Entertainment Law Journal*, 40(1), 81–112.
- Jessop, A., & Baker, T.A., III. (2019). Big data bust: Evaluating the risks of tracking NCAA athletes' biometric data. *Texas Review of Entertainment and Sports Law*, 20(1), 81–112.
- Jones, L., Konoval, T., & Toner, J. (2022). Sport and surveillance technologies. In Sanderson, J. (Ed.), *Sport, social media, and digital technology: Sociological approaches* (Vol. 15, pp. 165–183). Emerald Publishing Limited.
- Karkazis, K., & Fishman, J.R. (2017). Tracking US professional athletes: The ethics of biometric technologies. *The American Journal of Bioethics*, 17(1), 45–60. <https://doi.org/10.1080/15265161.2016.1251633>
- Keefer, Z. (2023). 'This kid's pretty special': Why Billy Napier thinks Anthony Richardson can make it. *The Athletic*. <https://www.nytimes.com/athletic/4317365/2023/03/17/anthony-richardson-billy-napier-nfl-draft/>
- Kincheloe, J.L., & McLaren, P. (2011). Rethinking critical theory and qualitative research. In J.L. Kincheloe (Ed.), *Key works in critical pedagogy* (pp. 285–326). Brill Academic Publishers.
- Kish, M. (2016). Nike's expanded effort to collect data from college athletes raises privacy concerns. *Portland Business Journal*. <https://www.bizjournals.com/portland/news/2016/09/15/nikes-expanded-effort-to-collect-data-from-college.html>
- Kluch, Y. (2023). Why don't more college athletes engage in activism? A multilevel analysis of barriers to activism in the hegemonic arena of intercollegiate sport. *Sport Management Review*, 26(5), 677–697. <https://doi.org/10.1080/14413523.2023.2175493>
- Livers v. NCAA. (2018). 17–4271 (United States District Court for the Eastern District of Pennsylvania 2018).
- Luczak, T., Burch, R., Lewis, E., Chander, H., & Ball, J. (2020). State-of-the-art review of athletic wearable technology: What 113 strength and conditioning coaches and athletic trainers from the USA said about technology in sports. *International Journal of Sports Science & Coaching*, 15(1), 26–40. <https://doi.org/10.1177/1747954119885244>
- Macaulay, C.D., Burton, L.J., & Woulfin, S. (2022). Making sense of competing logics in the collegiate athletic field: The sensemaking processes of college athletes. *Journal of Issues in Intercollegiate Athletics*, 15, 94–124.
- Makridis, C.A., & Thayer, J. (2023). Data is the new currency. *The Wall Street Journal*. <https://www.wsj.com/opinion/data-is-the-new-currency-big-tech-antitrust-free-services-platform-consumer-welfare-e5c74fb5>
- Mankin, J., Rivas, J., & Jewell, J. (2021). The effectiveness of college football recruiting ratings in predicting team success: A longitudinal study. *Research in Business and Economics Journal*, 14, 4–22.
- Mauthner, N.S., & Doucet, A. (2003). Reflexive accounts and accounts of reflexivity in qualitative data analysis. *Sociology*, 37(3), 413–431. <https://doi.org/10.1177/00380385030373002>
- McMahon, J.A., & Penney, D. (2013). (Self-) surveillance and (self-) regulation: Living by fat numbers within and beyond a sporting culture. *Qualitative Research in Sport, Exercise and Health*, 5(2), 157–178. <https://doi.org/10.1080/2159676X.2012.712998>
- McQuade, D. (2017). Why is this wearable-tech company helping college teams track how often athletes sleep, drink, and have sex? *Deadspin*. <https://deadspin.com/why-is-this-wearable-tech-company-helping-college-teams-1794218363/>
- Nakos, P. (2024). OneTeam Partners, CLC launch NIL opt-in program for EA Sports College Football 25. *On3*. <https://www.on3.com/nil/news/ea-sports-college-football-25-nil-opt-in-program-licensing-onetam-partners-clc-compass/>
- National Collegiate Athletic Association. (2023). NCAA committee on competitive safeguards and medical aspects of sports recommended statement on performance technologies. *NCAA*. https://ncaaorg.s3.amazonaws.com/committees/ncaa/safeguards/statements/Jun2023_CSMAS_PerformanceTechnologiesStatement.pdf
- National Collegiate Athletic Association. (n.d.). The 16 principles for conduct of intercollegiate athletics. *NCAA*. <https://www.ncaa.org/sports/2016/7/6/the-16-principles-for-conduct-of-intercollegiate-athletics.aspx>
- Nite, C., & Washington, M. (2024). Pollination, cultivation, and perennialization: Creating the amateur logic in collegiate sport. *Sport Management Review*, 28(1), 173–199. <https://doi.org/10.1080/14413523.2024.2403725>
- Novy-Williams, E. (2022). NCAA to allow sports betting data deals for schools, conferences. *Sportico*. <https://www.sportico.com/business/sports-betting/2022/ncaa-sports-betting-policy-1234673989/>
- Penn State Athletics. (2023). Corporate partners. *Penn State Athletic Department Website*. <https://gopsusports.com/sports/2018/8/8/sponsorship-psu-sponsorship.html.aspx>
- Rudner, J. (2022). Gators utilizing Catapult technology in recruiting. *247Sports*. <https://247sports.com/college/florida/Article/Florida-football-recruiting-Catapult-technology-Desirrio-Riles-Billy-Napier-Gators-2023-class-commitments-rankings-190754577/>
- Seshadri, D.R., Li, R.T., Voos, J.E., Rowbottom, J.R., Alfes, C.M., Zorman, C.A., & Drummond, C.K. (2019). Wearable sensors for monitoring the internal and external workload of the athlete. *NPJ Digital Medicine*, 2(1), 1–16. <https://doi.org/10.1038/s41746-019-0150-9>
- Smolenski, G. (2019). When the collection of biometric and performance data on college athletes goes too far. *Wake Forest Law Review*, 54(1), 279–301.
- Southall, R.M., Nagel, M.S., Staurowsky, E.J., Karcher, R.T., & Maxcy, J.G. (2023). *The NCAA and the exploitation of college profit-athletes: An amateurism that never was*. University of South Carolina Press.

- Southall, R.M., & Staurowsky, E.J. (2013). Cheering on the collegiate model: Creating, disseminating, and imbedding the NCAA's redefinition of amateurism. *Journal of Sport and Social Issues*, 37(4), 403–429. <https://doi.org/10.1177/0193723513498606>
- Southall, R.M., & Weiler, J.D. (2014). NCAA Division-I athletic departments: 21st century athletic company towns. *Journal of Issues in Intercollegiate Athletics*, 7, 161–186.
- Sparkes, A.C. (2002). *Telling tales in sport and physical activity: A qualitative journey*. Human Kinetics.
- Springer, D., Harry, M., & Heinrich, S. (2025). Investigating paternalistic attitudes in NCAA Division I Power 5 team conduct policies. *Innovative Higher Education*, 10, Article 9776. <https://doi.org/10.1007/s10755-024-09776-8>
- Steinbach, P. (2022). Penn State using wearables to aid athlete wellness. *Athletic Business*. <https://www.athleticbusiness.com/operations/media-technology/article/15295755/penn-state-using-wearables-to-aid-athlete-wellness>
- Stewart, J.J. (1987). The commodification of sport. *International Review for the Sociology of Sport*, 22(3), 171–192. <https://doi.org/10.1177/101269028702200303>
- Stokowski, S., Corr, C., Smith, K., Toney, E., Bostick, E.K., Russell, E., Prince, B., & Godfrey, M. (2025). “I didn’t come here to make money”: Exploring Power 4 athletes’ perceptions of exploitation. *Journal of Issues in Intercollegiate Athletics*, 18(1), 191–212. <https://doi.org/10.51221/sc.jiia.2025.18.1.10>
- Stokowski, S., Smith, A.B., Fridley, A., Corr, C., & Paule-Koba, A.L. (2023). “The most important shot you will ever take”: The burgeoning role of social media activism in challenging embedded NCAA patriarchy. *International Journal of Sport Communication*, 16(4), 482–492. <https://doi.org/10.1123/ijsc.2023-0159>
- Studnicka, A. (2020). The emergence of wearable technology and the legal implications for athletes, teams, leagues and other sports organizations across amateur and professional athletics. *DePaul Journal of Sports Law*, 16(1), 195–224.
- Thompson, C.J., Locander, W.B., & Pollio, H.R. (1989). Putting consumer experience back into consumer research: The philosophy and method of existential phenomenology. *Journal of Consumer Research*, 16(2), 133–146. <https://doi.org/10.1086/209203>
- Tracy, M. (2016). With wearable tech deals, new player data is up for grabs. *The New York Times*. <https://www.nytimes.com/2016/09/11/sports/ncaafootball/wearable-technology-nike-privacy-college-football.html>
- Vannini, C. (2025). College Football 25 broke video game records. Can EA Sports keep everyone happy? *The Athletic*. <https://www.nytimes.com/athletic/6161718/2025/02/27/ea-sports-college-football-26-video-game-sales/>
- WHOOOP. (2020). *NFLPA provides WHOOP to all active NFL players*. <https://www.whoop.com/us/en/press-center/nflpa-provides-whoop-to-players/>

Appendix: Interview Guide

Demographic Questions

- At what NCAA level do you currently compete?
- Do you compete in men's or women's collegiate athletics?
- What sport(s) do you compete in?

Experience with and Perceptions of Wearable Technology

- Are you aware that the NCAA permits the sale of college athletes' biometric data to organizations such as sportsbooks or apparel companies?
- How do you feel that the NCAA permits the sale of college athletes' biometric data to organizations such as sportsbooks or apparel companies?
- Does your university provide and utilize wearable technologies for athletes? What devices have been issued to you?
- Did you sign a waiver prior to using university-issued wearable technology?
- How were data from wearable technology used? Why were you told these data were collected?
- Do you believe college athletes should retain the rights to their biometric data?